

APPENDIX B - CORE SOLUTIONS

VERIFIED PROOFS

FAIL-CLOSED AUDITS

{{ SOLUTIONS\_DIRECTORY: Complete Verified Solutions for Appendix A Practice Items }}

SOLUTION: SOL-PRAC-01

TARGET: PRAC-01

{{ PRIMARY\_CAPABILITY: CAP-ASSIGN-OXIDATION-NUMBER-CORE }}

Reasoning Steps:

1. Identify compound type: Phosphoric acid ( $\text{H}_3\text{PO}_4$ ) is a neutral molecule (Net charge = 0).
2. Apply Rule Priority 4 (Hydrogen): Hydrogen is bonded to non-metals  $\Rightarrow \text{H} = +1$ .  
Total H contribution =  $3 \times (+1) = +3$ .
3. Apply Rule Priority 5 (Oxygen): Standard oxide bonding  $\Rightarrow \text{O} = -2$ . Total O =  $4 \times (-2) = -8$ .
4. Apply Priority 7 (Sum Rule): Let ON of P = x.  
 $(+3) + x + (-8) = 0 \Rightarrow x - 5 = 0 \Rightarrow x = +5$ .

{{ FINAL\_RESPONSE }}: The oxidation state of Phosphorus in  $\text{H}_3\text{PO}_4$  is +5.

{{ VERIFICATION\_STEP }}:  $(+3) + (+5) + (-8) = 0$ . Charge balance is exactly satisfied.

{{ CONDITION\_NOTE }}: Phosphorus uses all 5 valence electrons in bonding; +5 is its maximum state.

SOLUTION: SOL-PRAC-02

TARGET: PRAC-02

{{ PRIMARY\_CAPABILITY: CAP-TRACK-ELEMENT-CHARGE-CHANGE }}

Reasoning Steps:

1. Assign ON to every reactant species:
  - \* Free Zinc metal Zn(s): Elemental state  $\Rightarrow \text{ON} = 0$ .
  - \* In  $\text{HCl}(\text{aq})$ :  $\text{H} = +1$ ,  $\text{Cl} = -1$ .
2. Assign ON to every product species:
  - \* In  $\text{ZnCl}_2(\text{aq})$ : Group 12 metal Zn = +2; two chloride ions each  $\text{Cl} = -1$ .
  - \* In  $\text{H}_2(\text{g})$ : Free elemental diatomic hydrogen gas  $\Rightarrow \text{ON} = 0$ .
3. Track direction of electron transfer:
  - \* Zinc: Changes from 0 to +2  $\Rightarrow$  Loss of  $2\text{e}^- \Rightarrow$  UNDERGOES OXIDATION.
  - \* Hydrogen: Changes from +1 to 0  $\Rightarrow$  Gain of  $1\text{e}^-$  per atom  $\Rightarrow$  UNDERGOES REDUCTION.
  - \* Chloride: Stays at -1  $\Rightarrow$  Spectator ion (unchanged).

{{ FINAL\_RESPONSE }}: Zinc is oxidised (0  $\rightarrow$  +2); Hydrogen is reduced (+1  $\rightarrow$  0).

{{ VERIFICATION\_STEP }}: 1 mole Zn loses  $2\text{e}^-$ ; 2 moles  $\text{H}^+$  gain  $2(1\text{e}^-) = 2\text{e}^-$ . 100% conserved.

SOLUTION: SOL-PRAC-03

TARGET: PRAC-03

{{ PRIMARY\_CAPABILITY: CAP-ATTACH-REDOX-ROLES }}

Reasoning Steps:

1. Identify reacting species and changes in  $\text{Fe}_2\text{O}_3 + 3\text{CO} \rightarrow 2\text{Fe} + 3\text{CO}_2$ :
  - \* Iron in  $\text{Fe}_2\text{O}_3$ :  $2\text{Fe} + 3(-2) = 0 \Rightarrow \text{Fe} = +3$ . In product  $\text{Fe}(\text{s})$ :  $\text{Fe} = 0$ .  
Fe decreases from +3 to 0  $\Rightarrow$  Reduction. Therefore,  $\text{Fe}_2\text{O}_3$  is the OXIDISING AGENT.
  - \* Carbon in  $\text{CO}$ :  $\text{C} + (-2) = 0 \Rightarrow \text{C} = +2$ . In product  $\text{CO}_2$ :  $\text{C} + 2(-2) = 0 \Rightarrow \text{C} = +4$ .  
C increases from +2 to +4  $\Rightarrow$  Oxidation. Therefore,  $\text{CO}$  is the REDUCING AGENT.
2. Calculate electron count per formula unit of  $\text{Fe}_2\text{O}_3$ :
  - Each Iron atom decreases by 3 units:  $\text{Fe}^{3+} + 3\text{e}^- \rightarrow \text{Fe}^0$ .
  - 1 formula unit of  $\text{Fe}_2\text{O}_3$  contains 2 Iron atoms  $\Rightarrow 2 \times 3\text{e}^- = 6$  electrons accepted.
  - Check: 3 moles of  $\text{CO}$  each lose  $2\text{e}^- \Rightarrow 3 \times 2\text{e}^- = 6$  electrons donated. (Matches exactly).

{{ FINAL\_RESPONSE }}: Oxidising Agent =  $\text{Fe}_2\text{O}_3$ ; Reducing Agent =  $\text{CO}$ ; 6 moles  $\text{e}^-$  transferred.

{{ VERIFICATION\_STEP }}: Agent names attached to ENTIRE reactant species, not isolated atoms.

SOLUTION: SOL-PRAC-04

TARGET: PRAC-04

{{ PRIMARY\_CAPABILITY: CAP-DISCRIMINATE-PEROXIDE-EXCEPTION }}

Reasoning Steps (Causal Misconception Repair):

1. Physical impossibility check: Barium is an alkaline earth metal (Group 2,  $[\text{Xe}] 6\text{s}^2$ ).  
Its ionization energies allow only a +2 oxidation state (loss of two 6s electrons).  
Ba = +4 would require breaking into the stable closed noble-gas shell of Xenon (impossible).
2. Rule Priority Hierarchy: Priority 2 (Group 2 metal is ALWAYS +2) overrides Priority 5 (Oxygen).  
Therefore, Ba MUST be assigned +2 fixed.
3. Evaluate Oxygen via Sum Rule:
  - $(+2) + 2(\text{ON}_\text{O}) = 0 \Rightarrow 2(\text{ON}_\text{O}) = -2 \Rightarrow \text{ON of Oxygen} = -1$ .
4. Structural Confirmation:  $\text{BaO}_2$  contains the peroxide linkage  $[\text{O}-\text{O}]^{2-}$ .

{{ FINAL\_RESPONSE }}: Student claim is FALSE. Ba is fixed at +2; Oxygen is -1 (Peroxide).

{{ CONDITION\_NOTE }}: Group 1 and 2 metal fixed rules strictly precede the Oxygen default rule.